

Autogen_Multiagent_p4_h

March 22, 2026

1 Multi Agent : Autogen : StockAgent

SK

1. metrics (2 , SK)
2. (3) + (3) -> News Agent
3. Agent, Agent : metrics +
4. Decision Agent Agent Agent

1.0.1 Hugging Face Token

```
[29]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
[30]: %cd /content/drive/MyDrive/aiffel/3.rag/4
```

/content/drive/MyDrive/aiffel/3.rag/4

```
[2]: from dotenv import load_dotenv
import os

# load_dotenv("/content/drive/MyDrive/aiffel/env_keys/.env")
load_dotenv(".env")

print("HF_TOKEN loaded:", os.getenv("HF_TOKEN") is not None)
HF_TOKEN = os.getenv("HF_TOKEN")

print("OPENAI_API_KEY loaded:", os.getenv("OPENAI_API_KEY") is not None)
OPENAI_API_KEY = os.getenv("OPENAI_API_KEY")
```

HF_TOKEN loaded: True

OPENAI_API_KEY loaded: True

1.0.2

```
[1]: !pip install pykrx
```

Requirement already satisfied: pykrx in /usr/local/lib/python3.12/dist-packages (1.2.4)

Requirement already satisfied: requests>=2.32.0 in /usr/local/lib/python3.12/dist-packages (from pykrx) (2.32.4)

Requirement already satisfied: pandas<3.0,>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from pykrx) (2.2.2)

Requirement already satisfied: numpy<2.0,>=1.24.0 in /usr/local/lib/python3.12/dist-packages (from pykrx) (1.26.4)

Requirement already satisfied: deprecated>=1.2.14 in /usr/local/lib/python3.12/dist-packages (from pykrx) (1.3.1)

Requirement already satisfied: multipledispatch>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from pykrx) (1.0.0)

Requirement already satisfied: matplotlib>=3.8.0 in /usr/local/lib/python3.12/dist-packages (from pykrx) (3.10.0)

Requirement already satisfied: wrapt<3,>=1.10 in /usr/local/lib/python3.12/dist-packages (from deprecated>=1.2.14->pykrx) (2.1.2)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (4.62.1)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (1.5.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (26.0)

Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (11.3.0)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (3.3.2)

Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.8.0->pykrx) (2.9.0.post0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=2.2.0->pykrx) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas<3.0,>=2.2.0->pykrx) (2025.3)

Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.0->pykrx) (3.4.6)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.0->pykrx) (3.11)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.0->pykrx) (2.5.0)

Requirement already satisfied: certifi>=2017.4.17 in

```
/usr/local/lib/python3.12/dist-packages (from requests>=2.32.0->pykrx)
(2026.2.25)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.8.0->pykrx) (1.17.0)
```

```
[3]: from pykrx import stock
import pandas as pd
import numpy as np
```

```
[4]: def compute_metrics(df, price_col="Close", risk_free_rate=0.0, freq=252):
    """    df    ,    , MDD, Sharpe    ."""
    # 1.    →
    r = df[price_col].pct_change().dropna()

    # 2.    ,
    annual_return = r.mean() * freq
    annual_vol = r.std() * np.sqrt(freq)

    # 3.    (Maximum Drawdown, MDD)
    cum_ret = (1 + r).cumprod()
    cum_max = cum_ret.cummax()
    drawdown = (cum_ret - cum_max) / cum_max
    max_drawdown = drawdown.min()

    # 4. Sharpe ratio (    0    )
    excess_return = annual_return - risk_free_rate
    sharpe = excess_return / annual_vol if annual_vol > 1e-8 else np.nan

    # 5.    (CAGR)
    cagr = cum_ret.iloc[-1] ** (freq / len(r)) - 1

    # 6.
    metrics = {
        "annual_return": annual_return,
        "annual_volatility": annual_vol,
        "max_drawdown": max_drawdown,
        "sharpe_ratio": sharpe,
        "cagr": cagr,
        "n_days": len(r)
    }
    return pd.Series(metrics)
```

```
[5]: # 1)
def fetch_stock_history(ticker: str, years: int = 2) -> pd.DataFrame:
    end = pd.Timestamp.today().strftime("%Y%m%d")
    start = (pd.Timestamp.today() - pd.DateOffset(years=years)).
    ↪strftime("%Y%m%d")
```

```
df = stock.get_market_ohlcw(start, end, ticker)
return df
```

```
[6]: # 2)          compute_metrics
def compute_stock_metrics_for_ticker(ticker: str, years: int = 2) -> pd.Series:
    df = fetch_stock_history(ticker, years=years)
    # compute_metrics Close
    metrics = compute_metrics(df.rename(columns={" ": "Close"}))
    return metrics
```

```
[7]: # 3)          (decisionAgent  payload)
def compute_pair_metrics(t1: str = "005930", t2: str = "000660", years: int = 2):
    m1 = compute_stock_metrics_for_ticker(t1, years=years)
    m2 = compute_stock_metrics_for_ticker(t2, years=years)
    return {
        t1: m1.to_dict(),
        t2: m2.to_dict(),
    }
```

```
[ ]:
```

1.0.3 Step 0 :

Autogen Gemini API .

```
[7]: !pip install autogen
!pip install -U "autogen-agentchat"
!pip install "autogen-ext[openai]"
```

```
Collecting autogen
  Downloading autogen-0.11.4-py3-none-any.whl.metadata (24 kB)
Collecting ag2==0.11.4 (from autogen)
  Downloading ag2-0.11.4-py3-none-any.whl.metadata (38 kB)
Requirement already satisfied: anyio<5.0.0,>=4.0.0 in
/usr/local/lib/python3.12/dist-packages (from ag2==0.11.4->autogen) (4.12.1)
Collecting diskcache (from ag2==0.11.4->autogen)
  Downloading diskcache-5.6.3-py3-none-any.whl.metadata (20 kB)
Collecting docker (from ag2==0.11.4->autogen)
  Downloading docker-7.1.0-py3-none-any.whl.metadata (3.8 kB)
Collecting fast-depends<4.0.0,>=3.0.8 (from fast-
depends[pydantic]<4.0.0,>=3.0.8->ag2==0.11.4->autogen)
  Downloading fast_depends-3.0.8-py3-none-any.whl.metadata (7.7 kB)
Requirement already satisfied: httpx<1,>=0.28.1 in
/usr/local/lib/python3.12/dist-packages (from ag2==0.11.4->autogen) (0.28.1)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-
packages (from ag2==0.11.4->autogen) (26.0)
Requirement already satisfied: pydantic<3,>=2.6.1 in
```

```

/usr/local/lib/python3.12/dist-packages (from ag2==0.11.4->autogen) (2.12.3)
Requirement already satisfied: python-dotenv in /usr/local/lib/python3.12/dist-
packages (from ag2==0.11.4->autogen) (1.2.2)
Requirement already satisfied: termcolor in /usr/local/lib/python3.12/dist-
packages (from ag2==0.11.4->autogen) (3.3.0)
Requirement already satisfied: tiktoken in /usr/local/lib/python3.12/dist-
packages (from ag2==0.11.4->autogen) (0.12.0)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-
packages (from anyio<5.0.0,>=4.0.0->ag2==0.11.4->autogen) (3.11)
Requirement already satisfied: typing_extensions>=4.5 in
/usr/local/lib/python3.12/dist-packages (from
anyio<5.0.0,>=4.0.0->ag2==0.11.4->autogen) (4.15.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.28.1->ag2==0.11.4->autogen) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.28.1->ag2==0.11.4->autogen) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-
packages (from httpcore==1.*->httpx<1,>=0.28.1->ag2==0.11.4->autogen) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from
pydantic<3,>=2.6.1->ag2==0.11.4->autogen) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in
/usr/local/lib/python3.12/dist-packages (from
pydantic<3,>=2.6.1->ag2==0.11.4->autogen) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in
/usr/local/lib/python3.12/dist-packages (from
pydantic<3,>=2.6.1->ag2==0.11.4->autogen) (0.4.2)
Requirement already satisfied: requests>=2.26.0 in
/usr/local/lib/python3.12/dist-packages (from docker->ag2==0.11.4->autogen)
(2.32.4)
Requirement already satisfied: urllib3>=1.26.0 in
/usr/local/lib/python3.12/dist-packages (from docker->ag2==0.11.4->autogen)
(2.5.0)
Requirement already satisfied: regex>=2022.1.18 in
/usr/local/lib/python3.12/dist-packages (from tiktoken->ag2==0.11.4->autogen)
(2025.11.3)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests>=2.26.0->docker->ag2==0.11.4->autogen) (3.4.6)
Downloading autogen-0.11.4-py3-none-any.whl (13 kB)
Downloading ag2-0.11.4-py3-none-any.whl (1.1 MB)
1.1/1.1 MB
49.7 MB/s eta 0:00:00
Downloading fast_depends-3.0.8-py3-none-any.whl (25 kB)
Downloading diskcache-5.6.3-py3-none-any.whl (45 kB)
45.5/45.5 kB
5.4 MB/s eta 0:00:00
Downloading docker-7.1.0-py3-none-any.whl (147 kB)

```

147.8/147.8 kB

18.1 MB/s eta 0:00:00

Installing collected packages: diskcache, fast-depends, docker, ag2, autogen

Successfully installed ag2-0.11.4 autogen-0.11.4 diskcache-5.6.3 docker-7.1.0 fast-depends-3.0.8

Collecting autogen-agentchat

Downloading autogen_agentchat-0.7.5-py3-none-any.whl.metadata (2.5 kB)

Collecting autogen-core==0.7.5 (from autogen-agentchat)

Downloading autogen_core-0.7.5-py3-none-any.whl.metadata (2.3 kB)

Collecting jsonref~=1.1.0 (from autogen-core==0.7.5->autogen-agentchat)

Downloading jsonref-1.1.0-py3-none-any.whl.metadata (2.7 kB)

Requirement already satisfied: opentelemetry-api>=1.34.1 in /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-agentchat) (1.38.0)

Requirement already satisfied: pillow>=11.0.0 in /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-agentchat) (11.3.0)

Requirement already satisfied: protobuf~=5.29.3 in /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-agentchat) (5.29.6)

Requirement already satisfied: pydantic<3.0.0,>=2.10.0 in /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-agentchat) (2.12.3)

Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-agentchat) (4.15.0)

Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in /usr/local/lib/python3.12/dist-packages (from opentelemetry-api>=1.34.1->autogen-core==0.7.5->autogen-agentchat) (8.7.1)

Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.10.0->autogen-core==0.7.5->autogen-agentchat) (0.7.0)

Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.10.0->autogen-core==0.7.5->autogen-agentchat) (2.41.4)

Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.10.0->autogen-core==0.7.5->autogen-agentchat) (0.4.2)

Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.12/dist-packages (from importlib-metadata<8.8.0,>=6.0->opentelemetry-api>=1.34.1->autogen-core==0.7.5->autogen-agentchat) (3.23.0)

Downloading autogen_agentchat-0.7.5-py3-none-any.whl (119 kB)

119.3/119.3 kB

11.3 MB/s eta 0:00:00

Downloading autogen_core-0.7.5-py3-none-any.whl (101 kB)

101.9/101.9 kB

14.0 MB/s eta 0:00:00

Downloading jsonref-1.1.0-py3-none-any.whl (9.4 kB)

Installing collected packages: jsonref, autogen-core, autogen-agentchat
 Successfully installed autogen-agentchat-0.7.5 autogen-core-0.7.5 jsonref-1.1.0
 Collecting autogen-ext[openai]
 Downloading autogen_ext-0.7.5-py3-none-any.whl.metadata (7.3 kB)
 Requirement already satisfied: autogen-core==0.7.5 in
 /usr/local/lib/python3.12/dist-packages (from autogen-ext[openai]) (0.7.5)
 Requirement already satisfied: aiofiles in /usr/local/lib/python3.12/dist-
 packages (from autogen-ext[openai]) (24.1.0)
 Requirement already satisfied: openai>=1.93 in /usr/local/lib/python3.12/dist-
 packages (from autogen-ext[openai]) (2.28.0)
 Requirement already satisfied: tiktoken>=0.8.0 in
 /usr/local/lib/python3.12/dist-packages (from autogen-ext[openai]) (0.12.0)
 Requirement already satisfied: jsonref~=1.1.0 in /usr/local/lib/python3.12/dist-
 packages (from autogen-core==0.7.5->autogen-ext[openai]) (1.1.0)
 Requirement already satisfied: opentelemetry-api>=1.34.1 in
 /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-
 ext[openai]) (1.38.0)
 Requirement already satisfied: pillow>=11.0.0 in /usr/local/lib/python3.12/dist-
 packages (from autogen-core==0.7.5->autogen-ext[openai]) (11.3.0)
 Requirement already satisfied: protobuf~=5.29.3 in
 /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-
 ext[openai]) (5.29.6)
 Requirement already satisfied: pydantic<3.0.0,>=2.10.0 in
 /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-
 ext[openai]) (2.12.3)
 Requirement already satisfied: typing-extensions>=4.0.0 in
 /usr/local/lib/python3.12/dist-packages (from autogen-core==0.7.5->autogen-
 ext[openai]) (4.15.0)
 Requirement already satisfied: anyio<5,>=3.5.0 in
 /usr/local/lib/python3.12/dist-packages (from openai>=1.93->autogen-ext[openai])
 (4.12.1)
 Requirement already satisfied: distro<2,>=1.7.0 in
 /usr/local/lib/python3.12/dist-packages (from openai>=1.93->autogen-ext[openai])
 (1.9.0)
 Requirement already satisfied: httpx<1,>=0.23.0 in
 /usr/local/lib/python3.12/dist-packages (from openai>=1.93->autogen-ext[openai])
 (0.28.1)
 Requirement already satisfied: jiter<1,>=0.10.0 in
 /usr/local/lib/python3.12/dist-packages (from openai>=1.93->autogen-ext[openai])
 (0.13.0)
 Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-
 packages (from openai>=1.93->autogen-ext[openai]) (1.3.1)
 Requirement already satisfied: tqdm>4 in /usr/local/lib/python3.12/dist-packages
 (from openai>=1.93->autogen-ext[openai]) (4.67.3)
 Requirement already satisfied: regex>=2022.1.18 in
 /usr/local/lib/python3.12/dist-packages (from tiktoken>=0.8.0->autogen-
 ext[openai]) (2025.11.3)
 Requirement already satisfied: requests>=2.26.0 in

```

/usr/local/lib/python3.12/dist-packages (from tiktoken>=0.8.0->autogen-
ext[openai]) (2.32.4)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-
packages (from anyio<5,>=3.5.0->openai>=1.93->autogen-ext[openai]) (3.11)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->openai>=1.93->autogen-ext[openai]) (2026.2.25)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-
packages (from httpx<1,>=0.23.0->openai>=1.93->autogen-ext[openai]) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-
packages (from httpcore==1.*->httpx<1,>=0.23.0->openai>=1.93->autogen-
ext[openai]) (0.16.0)
Requirement already satisfied: importlib-metadata<8.8.0,>=6.0 in
/usr/local/lib/python3.12/dist-packages (from opentelemetry-
api>=1.34.1->autogen-core==0.7.5->autogen-ext[openai]) (8.7.1)
Requirement already satisfied: annotated-types>=0.6.0 in
/usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.10.0->autogen-
core==0.7.5->autogen-ext[openai]) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in
/usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.10.0->autogen-
core==0.7.5->autogen-ext[openai]) (2.41.4)
Requirement already satisfied: typing-inspection>=0.4.2 in
/usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.10.0->autogen-
core==0.7.5->autogen-ext[openai]) (0.4.2)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests>=2.26.0->tiktoken>=0.8.0->autogen-ext[openai]) (3.4.6)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests>=2.26.0->tiktoken>=0.8.0->autogen-ext[openai]) (2.5.0)
Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.12/dist-
packages (from importlib-metadata<8.8.0,>=6.0->opentelemetry-
api>=1.34.1->autogen-core==0.7.5->autogen-ext[openai]) (3.23.0)
Downloading autogen_ext-0.7.5-py3-none-any.whl (331 kB)
331.4/331.4 kB
28.6 MB/s eta 0:00:00
Installing collected packages: autogen-ext
Successfully installed autogen-ext-0.7.5

```

```
[ ]: # from google.colab import userdata
```

```
# GOOGLE_API_KEY = userdata.get('GEMINI_KEY')
```

```
[8]: from autogen_agentchat.agents import AssistantAgent
      from autogen_agentchat.conditions import TextMentionTermination
      from autogen_ext.models.openai import OpenAIChatCompletionClient
```


1.0.4 Step 1 :

description , !

1.0.5 Q1.

Hugging Face LLM

```
[27]: !pip install transformers accelerate sentencepiece -q
```

```
[9]: #

import torch
from transformers import pipeline, AutoTokenizer, AutoModelForCausalLM

#
# MODEL_NAME = "beomi/Llama-3-Open-Ko-8B"
# MODEL_NAME = "MLP-KTLim/llama-3-Korean-Bllossom-8B"
MODEL_NAME = "Bllossom/llama-3.2-Korean-Bllossom-3B"

tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)

model = AutoModelForCausalLM.from_pretrained(
    MODEL_NAME,
    torch_dtype=torch.bfloat16,
    device_map="auto",          # GPU
)

hf_generator = pipeline(
    "text-generation",
    model=model,
    tokenizer=tokenizer,
)

# hf_generator = pipeline(
#     "text-generation",
#     model=model,
#     tokenizer=tokenizer,
#     # device              (device_map    )
#     max_new_tokens=128,      #              256 -> 128
#     do_sample=True,
#     temperature=0.7,
#     top_p=0.9,
# )
```

```
config.json: 0%|          | 0.00/904 [00:00<?, ?B/s]
```

```
C:\Users\User\.conda\envs\aiifel_s\Lib\site-
```

packages\huggingface_hub\file_download.py:129: UserWarning: `huggingface\_hub` cache-system uses symlinks by default to efficiently store duplicated files but your machine does not support them in
C:\Users\User\.cache\huggingface\hub\models--Bllossom--llama-3.2-Korean-Bllossom-3B. Caching files will still work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF\_HUB\_DISABLE\_SYMLINKS\_WARNING` environment variable. For more details, see https://huggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: <https://docs.microsoft.com/en-us/windows/apps/get-started/enable-your-device-for-development>

```
warnings.warn(message)
```

tokenizer_config.json: 0.00B [00:00, ?B/s]

tokenizer.json: 0%| | 0.00/17.2M [00:00<?, ?B/s]

special_tokens_map.json: 0%| | 0.00/296 [00:00<?, ?B/s]

`torch\_dtype` is deprecated! Use `dtype` instead!

model.safetensors.index.json: 0.00B [00:00, ?B/s]

Downloading (incomplete total...): 0.00B [00:00, ?B/s]

Fetching 2 files: 0%| | 0/2 [00:00<?, ?it/s]

Loading weights: 0%| | 0/254 [00:00<?, ?it/s]

generation_config.json: 0%| | 0.00/180 [00:00<?, ?B/s]

```
[10]: #
out = hf_generator(
    " . .",
    max_new_tokens=80,
    do_sample=True,
    temperature=0.3,
    top_p=0.9,
    return_full_text=False,
)
print(out[0]["generated_text"])
```

Passing `generation\_config` together with generation-related arguments=({'top_p', 'max_new_tokens', 'temperature', 'do_sample'}) is deprecated and will be removed in future versions. Please pass either a `generation\_config` object OR all generation parameters explicitly, but not both.

Setting `pad\_token\_id` to `eos\_token\_id`:128001 for open-end generation. Both `max\_new\_tokens` (=80) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information.

(https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

TV, , , , ,

```
[41]: # import torch
# from transformers import pipeline, AutoTokenizer, AutoModelForCausalLM

# #
# # MODEL_NAME = "beomi/Llama-3-Open-Ko-8B"
# MODEL_NAME = "MLP-KTLim/llama-3-Korean-Bllossom-8B"

# #
# # MODEL_NAME = "meta-llama/Meta-Llama-3-8B-Instruct"

# tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME,
#                                           token=HF_TOKEN)

# model = AutoModelForCausalLM.from_pretrained(
#     MODEL_NAME,
#     torch_dtype=torch.bfloat16,
#     device_map="auto",          # GPU
#     token=HF_TOKEN,
# )

# generator = pipeline(
#     "text-generation",
#     model=model,
#     tokenizer=tokenizer,
# )
```

```
[11]: class LocalHFClient:
    def __init__(self, generator):
        self.generator = generator

    async def generate(self, prompt: str, max_new_tokens: int = 512) -> str:
        # async
        # ( ThreadPoolExecutor blocking )
        outputs = self.generator(
            prompt,
            max_new_tokens=max_new_tokens,
            return_full_text=False,
        )
        return outputs[0]["generated_text"]
```

```
[12]: client_small = LocalHFClient(hf_generator)
```

1.0.6 Q2.

!

scenario writer, scene planner, storyboard artist

```
[13]: SAMSUNG_ANALYST_PROMPT = """
      (005930) .

      :
1)      2      (metrics)
2)

metrics annual_return, annual_volatility, max_drawdown,
sharpe_ratio, cagr, n_days .

:
-      ,      .
-      ,      .

      (      ):

[1]      (3~5 )
-      (annual_return, cagr),      (annual_volatility),      (max_drawdown),
      (      , sharpe_ratio)      .
-      ,      2      ( / )      .

[2]      .      (3~5 )
-      /      ,
      . ( : HBM,      ,      ,      )
-      ,      .

[3] 3~6      (2~3 )
-      : ( / / )      .
-
-      .

      :
-      .
-      , / / /      ,      .
-      [1]~[3]      .
""" .strip()

HYNIX_ANALYST_PROMPT = """
      SK      (000660) .

      :
1)      2      (metrics)
```

```

2)          SK

metrics  annual_return, annual_volatility, max_drawdown,
sharpe_ratio, cagr, n_days

:
- SK          ,          .
-          ,          .

(          ):

[1]          (3~5 )
- (annual_return, cagr), (annual_volatility), (max_drawdown),
  ( , sharpe_ratio) SK .

[2] . (3~5 )
- / ,
  . (: HBM, D , , . , )

[3] 3~6 (2~3 )
- : ( / / ) .
- .

:
- .
- SK , / / / , , .
- [1]~[3] .
"""".strip()

```

```

[14]: async def run_samsung_analyst(prompt_body: str) -> str:
    # prompt_body: build_analyst_prompt_from_news_report
    full_prompt = f"{SAMSUNG_ANALYST_PROMPT}\n\n[ ]\n{prompt_body}"
    return await client_small.generate(full_prompt, max_new_tokens=512)

async def run_hynix_analyst(prompt_body: str) -> str:
    full_prompt = f"{HYNIX_ANALYST_PROMPT}\n\n[ ]\n{prompt_body}"
    return await client_small.generate(full_prompt, max_new_tokens=512)

```

```

[15]: DECISION_SYSTEM_PROMPT = """

(005930) SK (000660) (metrics)
,

:
- (005930) metrics
- SK (000660) metrics
metrics  annual_return, annual_volatility, max_drawdown,

```

```

sharpe_ratio, cagr, n_days      .

:
-      (005930, 000660)      . .
-      ,      .

      (      ):

1)
- = , = (005930) / SK (000660)      1      .
-      .
      ( : "0.25 ( 2      )" )

2)
-      2~3 ( )
- SK      2~3 ( )
-      ,      /      .

3)      (3~6 )
-      ,
      3~4      .
-      2~3      .
-      " / / "      ,
      .

:
-      .
-      SK      , / / / , ,      .
-      1)~3)      .
"""".strip()

```

1.0.7 ->

```

[16]: NEWS_SYSTEM_PROMPT = ""

:
- ( : 005930, 000660)
-
- 1~3      (      )

:
-      ,
-      ( : [ 2], [ 3] )      .
-      .
-      " : " "[ 1]"      ,      .

```

```

( ):

[1] (2~3 )
- .

[2] 1~3 ( 3~5 )
- , .

[3] ( 2~3 )
- : .
- : .

[4] (1 , 3~4 )
- ( / / ) , .

:
- .
- , , , , , .
- , , , , , .
- , [1]~[4] .
""".strip()

```

```

[17]: async def get_news_report_from_manual_input(
    ticker: str,
    company_name: str,
    news_items: list[dict], # {"title": str, "body": str}
) -> str:
    joined_texts = "\n\n".join(
        f"[ {i}]\n : {item['title']}\n :\n{item['body']}"
        for i, item in enumerate(news_items, start=1)
    )

    prompt = f"""{NEWS_SYSTEM_PROMPT}

: {ticker}
: {company_name}

[ 1~3 ]
{company_name} .

:
{joined_texts}
"""

    report = await client_small.generate(
        prompt,

```

```

        max_new_tokens=200,
    )
    return report

```

```

[18]: def build_analyst_prompt_from_news_report(
    ticker: str,
    metrics: dict,
    news_report: str,
) -> str:
    # metrics
    metrics_lines = "\n".join(
        f"- {k}: {v}" for k, v in metrics.items()
    )

    return (
        f" : {ticker}\n"
        f"    (metrics):\n{metrics_lines}\n\n"
        f"    NewsAgent      1~3      .\n"
        f"    ,                  .\n\n"
        f"[      ]\n{news_report}\n"
    )

```

```
[ ]:
```

```
[ ]:
```

1.0.8 Q4.

```

!
result

```

```

[19]: async def run_decision_agent(
    metrics_dict,
    samsung_news_report: str,
    hynix_news_report: str,
    samsung_report: str,
    hynix_report: str,
) -> str:
    metrics_005930 = metrics_dict["005930"]
    metrics_000660 = metrics_dict["000660"]

    decision_prompt = f"""{DECISION_SYSTEM_PROMPT}

[ (005930) metrics]
{metrics_005930}

[ (000660) metrics]
{metrics_000660}

```



```

[ (NewsAgent)]
{samsung_news_report}

[ (NewsAgent)]
{hynix_news_report}

[ ]
{samsung_report}

[ ]
{hynix_report}
"""

# client_decision HuggingFace/ LLM
return await client_small.generate(decision_prompt, max_new_tokens=1024)

```

```

[20]: # 3

samsung_sample_news = [
    {
        "title": " , HBM3E ... ",
        "body": (
            " HBM3E . "
            " AI GPU , "
            "AI . "
            " HBM3E HBM , "
            " HBM4 ."
        ),
    },
    {
        "title": " , 2024 ... ",
        "body": (
            " 2024 , "
            " . "
            " , "
            " . "
            " , "
            " ."
        ),
    },
    {
        "title": " ",
        "body": (
            " AI , "
            " . "
            " HBM D "

```

```

        "
        "
        "
        "
    ),
},
]

```

```

[21]: # 3

skhynix_sample_news = [
    {
        "title": "SK , HBM3E ·HBM4 ... AI ",
        "body": (
            "SK HBM3E "
            "AI . "
            " , , AWS , "
            " HBM4 . "
            " SK HBM3E HBM4 "
            " 2~3 ."
        ),
    },
    {
        "title": "SK , ... . ",
        "body": (
            "SK 2025~2027 , "
            " "
            " . "
            " HBM "
            " . "
            " SK "
            " ."
        ),
    },
    {
        "title": " , SK ... 90 ",
        "body": (
            " AI "
            " SK . "
            " 2 , "
            " SK 90 . "
            " 130 ~140 , "
            "HBM . "
            " "
            " ."
        ),
    },
},

```

```
]
```

1.0.9 ()

```
[13]: test_prompt = " . ' ' ."  
  
test_report = await client_small.generate(  
    test_prompt,  
    max_new_tokens=50,  
)  
print("=== TEST OUTPUT ===")  
print(test_report)
```

Passing `generation_config` together with generation-related arguments=(`{'max_new_tokens'}`) is deprecated and will be removed in future versions. Please pass either a `generation_config` object OR all generation parameters explicitly, but not both.
Both `max_new_tokens` (=50) and `max_length` (=4096) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information.

(https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

```
=== TEST OUTPUT ===  
' , . ' . ( : ,  
. ) 2023 3
```

```
[26]: prompt = f"{NEWS_SYSTEM_PROMPT}"  
  
: 005930  
:  
  
[ ]  
 . ,  
 .  
  
===  
[ 1]  
: {samsung_sample_news[0]['title']}  
:  
{samsung_sample_news[0]['body']}  
===  
""  
  
report = await client_small.generate(  
    prompt,  
    max_new_tokens=200,  
)  
print("=== MODEL OUTPUT START ===")
```

```
print(report)
print("=== MODEL OUTPUT END ===")
```

Both `max\_new\_tokens` (=200) and `max\_length` (=4096) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information.

(https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

```
=== MODEL OUTPUT START ===
```

```
[ 2]
:   , 5G          40% ...
:
    5G          .          ,          5G          40%
    ,          .          5G
    ,          .
```

```
===
```

```
[ 3]
:   , AI·IoT      ...
:
```

```
=== MODEL OUTPUT END ===
```

```
[22]: import time

async def run_full_pipeline_with_news_agent(years: int = 2):
    t0 = time.time()

    # 1)
    print("start metrics", flush=True)
    metrics_dict = compute_pair_metrics("005930", "000660", years=years)
    metrics_005930 = metrics_dict["005930"]
    metrics_000660 = metrics_dict["000660"]
    print("done metrics", time.time() - t0, flush=True)

    # 2) NewsAgent
    print("start samsung news", flush=True)
    samsung_news_report = await get_news_report_from_manual_input("005930",
↪ " ", samsung_sample_news)
    print("done samsung news", time.time() - t0, flush=True)

    # print("=== SAMSUNG NEWS REPORT ===")
    # print(samsung_news_report)

    print("start hynix news", flush=True)
    hynix_news_report = await get_news_report_from_manual_input("000660",
↪ "SK ", skhynix_sample_news)
    print("done hynix news", time.time() - t0, flush=True)
```

```

# print("\n=== SK HYNIX NEWS REPORT ===")
# print(hynix_news_report)

# 3)          ( + NewsAgent )
samsung_prompt = build_analyst_prompt_from_news_report(
    "005930",
    metrics_005930,
    samsung_news_report,
)
hynix_prompt = build_analyst_prompt_from_news_report(
    "000660",
    metrics_000660,
    hynix_news_report,
)

print("start samsung analyst", flush=True)
samsung_report = await run_samsung_analyst(samsung_prompt)
print("done samsung analyst", time.time() - t0, flush=True)
# await asyncio.sleep(1.5)

print("start hynix analyst", flush=True)
hynix_report = await run_hynix_analyst(hynix_prompt)
print("done hynix analyst", time.time() - t0, flush=True)
# await asyncio.sleep(1.5)

# 4) DecisionAgent

print("start decision", flush=True)
final_decision = await run_decision_agent(
    metrics_dict, samsung_news_report, hynix_news_report,
    samsung_report, hynix_report,
)
print("done decision", time.time() - t0, flush=True)

return {
    "metrics": metrics_dict,
    "samsung_news_report": samsung_news_report,
    "hynix_news_report": hynix_news_report,
    "samsung_report": samsung_report,
    "hynix_report": hynix_report,
    "final_decision": final_decision,
}

```

```

[23]: result = await run_full_pipeline_with_news_agent(years=2)

print("=== SAMSUNG NEWS (NewsAgent) ===")
print(result["samsung_news_report"])

```

```

print("\n=== HYNIX NEWS (NewsAgent) ===")
print(result["hynix_news_report"])

print("\n=== SAMSUNG ANALYST REPORT ===")
print(result["samsung_report"])

print("\n=== HYNIX ANALYST REPORT ===")
print(result["hynix_report"])

print("\n=== FINAL DECISION ===")
print(result["final_decision"])

```

start metrics

done metrics 0.062204837799072266

start samsung news

Passing `generation\_config` together with generation-related arguments=({'max_new_tokens'}) is deprecated and will be removed in future versions. Please pass either a `generation\_config` object OR all generation parameters explicitly, but not both.

Setting `pad\_token\_id` to `eos\_token\_id`:128001 for open-end generation.

Both `max\_new\_tokens` (=200) and `max\_length` (=20) seem to have been set.

`max\_new\_tokens` will take precedence. Please refer to the documentation for more information.

(https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

done samsung news 8.423295974731445

start hynix news

Setting `pad\_token\_id` to `eos\_token\_id`:128001 for open-end generation.

Both `max\_new\_tokens` (=200) and `max\_length` (=20) seem to have been set.

`max\_new\_tokens` will take precedence. Please refer to the documentation for more information.

(https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

done hynix news 16.762725353240967

start samsung analyst

Setting `pad\_token\_id` to `eos\_token\_id`:128001 for open-end generation.

Both `max\_new\_tokens` (=512) and `max\_length` (=20) seem to have been set.

`max\_new\_tokens` will take precedence. Please refer to the documentation for more information.

(https://huggingface.co/docs/transformers/main/en/main_classes/text_generation)

done samsung analyst 36.158782958984375

start hynix analyst

Setting `pad\_token\_id` to `eos\_token\_id`:128001 for open-end generation.

Both `max\_new\_tokens` (=512) and `max\_length` (=20) seem to have been set.

`max\_new\_tokens` will take precedence. Please refer to the documentation for

more information.
https://huggingface.co/docs/transformers/main/en/main_classes/text_generation
done hynix analyst 55.25229525566101
start decision
Setting `pad\_token\_id` to `eos\_token\_id`:128001 for open-end generation.
Both `max\_new\_tokens` (=1024) and `max\_length` (=20) seem to have been set.
`max\_new\_tokens` will take precedence. Please refer to the documentation for
more information.
https://huggingface.co/docs/transformers/main/en/main_classes/text_generation
done decision 113.35283017158508
=== SAMSUNG NEWS (NewsAgent) ===

[1]
, (memory, logic,
system LSI), (OLED, LCD), (TV, ,) .

[2] 1~3
- , HBM3E ...
- , 2024 ...
-

[3]
:
- HBM3E

=== HYNIX NEWS (NewsAgent) ===
- [1]
SK , HBM3E HBM4 AI .
, , . AI
.
- [2] 1~3
* SK HBM3E AI .
* ,

=== SAMSUNG ANALYST REPORT ===
-
-
:
-
-

[1] (3~5)
2 , (annual_return, cagr) 2 62.4%
, (annual_volatility) (max_drawdown) . ,

(, sharpe_ratio) 1.40 , .
2 .

[2] . (3~5)
HBM3E , , . HBM3E
 , . , .

[3] 3~6 (2~3)
:
: HBM3E , .
HBM3E , .

=== HYNIX ANALYST REPORT ===

* Server D , .

[1] (3~5)
SK (annual_return, cagr)
 . (annual_volatility) relative investors
 . (max_drawdown) 0.365% . (,
sharpe_ratio) 1.92 , SK .

[2] . (3~5)
SK HBM3E AI , .
 , . Server D
 . SK .

[3] 3~6 (2~3)
:
:
- (annual_return, cagr) .
- , .
- Server D . SK

=== FINAL DECISION ===

:
-
-
[(005930) metrics]
{'annual_return': 0.5666509460613119, 'annual_volatility':
0.40458214147709376, 'max_drawdown': -0.43166287015945304, 'sharpe_ratio':
1.400583189343255, 'cagr': 0.6237274811659408, 'n_days': 482.0}

[(000660) metrics]
{'annual_return': 1.0940883755804756, 'annual_volatility':


```
0.5696343240109262, 'max_drawdown': -0.36597510373443987, 'sharpe_ratio':  
1.9206854809533735, 'cagr': 1.5362325140788702, 'n_days': 482.0}
```

```
[1]
```

```
|      |      (005930) | SK   (000660) |  
| :----- | :----- | :----- |  
| Annual Return | 0.5666509460613119 | 1.0940883755804756 |  
| Annual Volatility | 0.40458214147709376 | 0.5696343240109262 |  
| Max Drawdown | -0.43166287015945304 | -0.36597510373443987 |  
| Sharpe Ratio | 1.400583189343255 | 1.9206854809533735 |  
| CAGR | 0.6237274811659408 | 1.5362325140788702 |  
| n_days | 482.0 | 482.0 |
```

```
[2]
```

```
-  
-
```

```
SK  
- HBM3E  
-  
- Server D
```

```
[3]
```

```
SK      HBM3E      ,      .  
      ,      ,  
      .      SK  
      .  
SK      .      .
```

```
[ ]: 
```

1.0.10

- , :
— , .
— , .
- :
—
- . :
— , .
- . :
—

```
[ ]: 
```

```
[ ]: 
```

[]: